

From business objective to priced decision

BUSINESS OBJECTIVE

\$100M

Incremental revenue from international products

ANALYTICAL QUESTIONS

- Q1 Can we close the revenue gap by changing price?
- Q2 How would different customer segments be affected by a price change?

HOW AN ACCOUNT CAN RESPOND TO A PRICE CHANGE

Leaves Switches carrier. Costs the whole account, not just the pass.	Substitutes Stops using the pass in favor of eSIM, local SIM or Wi-Fi.	Reduces Keeps the pass but buys fewer days or activates fewer lines.	Unchanged Same usage pattern at the new price.	Increases Buys more days or lines, typically in response to a lower price.
Revenue per trip = (\$10 + \$5 × (lines – 1)) × days		Leaves, substitutes and unchanged each imply a determined revenue outcome. Reduces and increases do not: the state gives the direction, and lines and days give the size.		

WHAT THE MODELS LEARN FROM

Travel and usage history When accounts traveled, where, for how long, and how much of the trip the pass covered.	Account structure Lines on the account, which of them traveled, plan and tenure.	Observed price variation Periods where the price an account faced actually differed. Without variation there is nothing to estimate a response from.	Market context International travel volume and share, used to set the base the price acts on.
Note on data: the underlying sources, field definitions and results are proprietary to AT&T. Data is described here by category only, and every figure shown is a reasonable approximation presented to illustrate the method rather than actual company performance.			

THE PRICE RANGE UNDER TEST

–10% to +20%	Every scenario is simulated across this range. The +20% ceiling is a leadership constraint, not a modeled optimum. If revenue is still climbing at the cap, the analysis reports a bounded answer rather than the unconstrained one.
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HOW THE QUESTIONS GET ANSWERED · UNIT OF ANALYSIS: THE ACCOUNT-TRIP

STAGE A · WHICH STATE

Predict the response

A multiclass model trained on past price movement, learning how account characteristics predict which state an account lands in at a given price.

Output: a probability across all five states, summing to 1, per account-trip.

STAGE B · HOW MUCH

Size the movement

Reduces and increases need a magnitude, fit only on accounts in that state. Nine days instead of ten is not two days instead of ten.

Output: predicted lines and days, conditional on the state.

STAGE C · ANSWERS Q1

Simulate 12 months

Monte Carlo simulation draws each account's state and magnitude across a year of trips, repeated at every price point in the tested range.

Output: a distribution of 12-month revenue outcomes, not a point estimate.

WHAT THE SIMULATION VARIES

Behavioral uncertainty State drawn per trip from the predicted probabilities, magnitude drawn from its conditional distribution.	International travel volume How much travel happens at all, driven by economic expansion and contraction and by seasonality across the year.	Market share What portion of those travelers are ours to price to, itself moving over the horizon.
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Travel volume and market share set the base the price acts on, and both are uncertain in their own right. Separating them from behavioral response matters: a downside outcome caused by a contracting travel market is a different decision from one caused by customers rejecting the price.

ANSWERS Q2 · SEGMENTATION AND PROFILING

SEGMENT

Group on the probability vector

Accounts are clustered on their full distribution across the five states, not on a single sensitivity score. An account carrying 40% substitute risk is a different problem from one carrying 40% reduce risk, even when expected revenue matches.

Output: segments that differ in how they respond, not only how much.

PROFILE

Describe each group in business terms

The model features are read back across each cluster to characterize it: travel frequency, destination mix, account size, tenure and spend. Profiling does not create the segment, it explains one that already exists.

Output: named segments a product owner can price to.

Each segment carries its own simulated revenue curve across the price range, so the two questions resolve together: whether the gap can be closed, and which customers close it.

BEFORE IMPLEMENTATION

Six definitions the plan depends on. Each one changes what gets built, and two of them can invalidate the analysis without ever producing an error.

Label construction How a historical account-trip is assigned one of the five states. Reduce versus unchanged needs a baseline and a threshold. Substitution is unobserved: absence of purchase and absence of need look identical in the data. Whatever rule is written here is what the model learns.	Simulation correlation Accounts cannot be drawn independently. Macro conditions move everyone at once, so draws are correlated. Independent draws produce an unrealistically narrow revenue distribution and report false confidence.
Trip definition Where one trip ends and the next begins: a gap in network presence, a country change, a calendar rule. This sets the row grain and every feature built on it.	Class imbalance Leaving is rare and the most expensive state. Untreated, the model predicts it almost never and the simulation understates the downside.
Temporal validation and calibration Split by time, never at random. The simulation consumes probabilities as truth, so they have to be calibration-checked rather than only scored for accuracy.	Acceptance criteria What result is strong enough to act on, agreed before any results exist.

DELIVERY · PUTTING THE MODEL IN THE HANDS OF THE BUSINESS

Power BI scenario and sensitivity tool

Simulation results are published to Power BI so business leaders can run their own scenarios against the final model rather than requesting a new analysis for every question.

Set the price anywhere in the –10% to +20% range and see the revenue distribution move.

Switch the macro scenario between expansion, base and contraction.

Filter to a segment to see who carries the upside and who carries the risk.

Read the range, not a point: outcomes are shown with their downside, not as a single number.